Statistical Analysis Report

Analysis Summary Template

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1 Summary

Provide a brief overview of the analysis objectives, key findings, and main conclusions. This section should be concise and accessible to non-technical stakeholders. Max 1 paragraph, no bullet points

2 Introduction

This analysis is to demonstrate the use of Artificial Intelligence (AI) Large Language Models (LLMs)to analyse data. The goal here is to stress test LLMs on a simple analysis task that involves a linear mixed effects model with a binary dependent variable and a blocking factor. The report will be in the form of a statistical analysis report.

3 Data

The data are from a subset of the Chz2006 dataset in the R package {agricolae} (Mendiburu and Yaseen 2020). There are 3 factors of interest(crop, amendment, and block) and a binary dependent variable (wilt_percent). There is also a plant individual factor which we will not consider here. There are n = 1920 observations in the data. The data experimentally meassures the incidence and performance of healthy tubers and rotten potato field infested with naturally Ralstonia solanacearum Race 3/Bv 2A, after application of inorganic amendments and a rotation crop in Carhuaz Peru, 2006.

```
'data.frame':   1920 obs. of  6 variables:
 $ amendment   : chr  "0C" "0C" "0C" "0C" ...
 $ crop        : chr  "Pea" "Pea" "Pea" "Pea" ...
 $ block       : Factor w/ 3 levels "1","2","3": 1 1 1 1 1 1 1 1 1 1 ...
 $ plant       : int  1 2 3 4 5 6 7 8 9 10 ...
 $ wilt_percent: int  0 0 50 0 0 0 0 0 0 0 ...
 $ wilt_bin    : num  0 0 1 0 0 0 0 0 0 0 ...
```

Table 1: Dataset sample

amendment	crop	block	plant	wilt_percent	wilt_bin
0C	Pea	1	1	0	0
0C	Pea	1	2	0	0
0C	Pea	1	3	50	1
0C	Pea	1	4	0	0
0C	Pea	1	5	0	0
0C	Pea	1	6	0	0

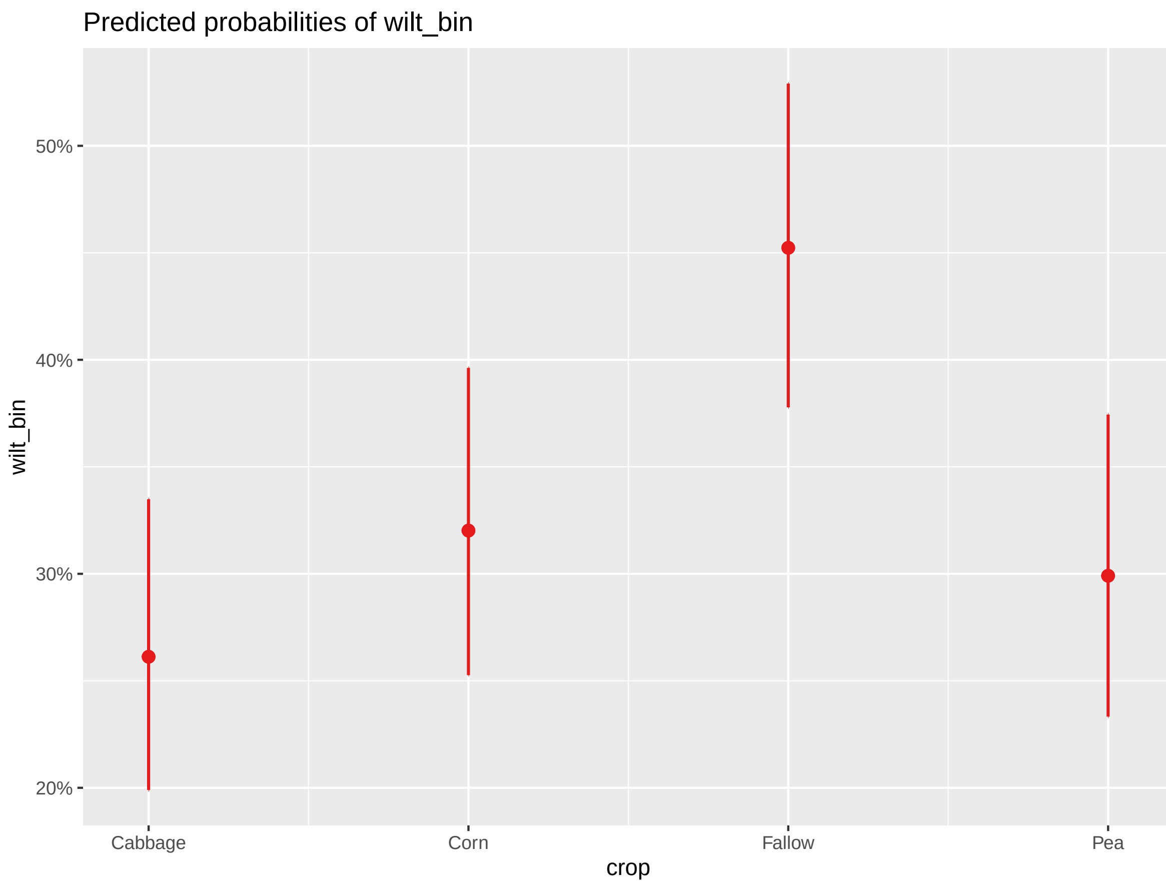
4 Statistical Analysis

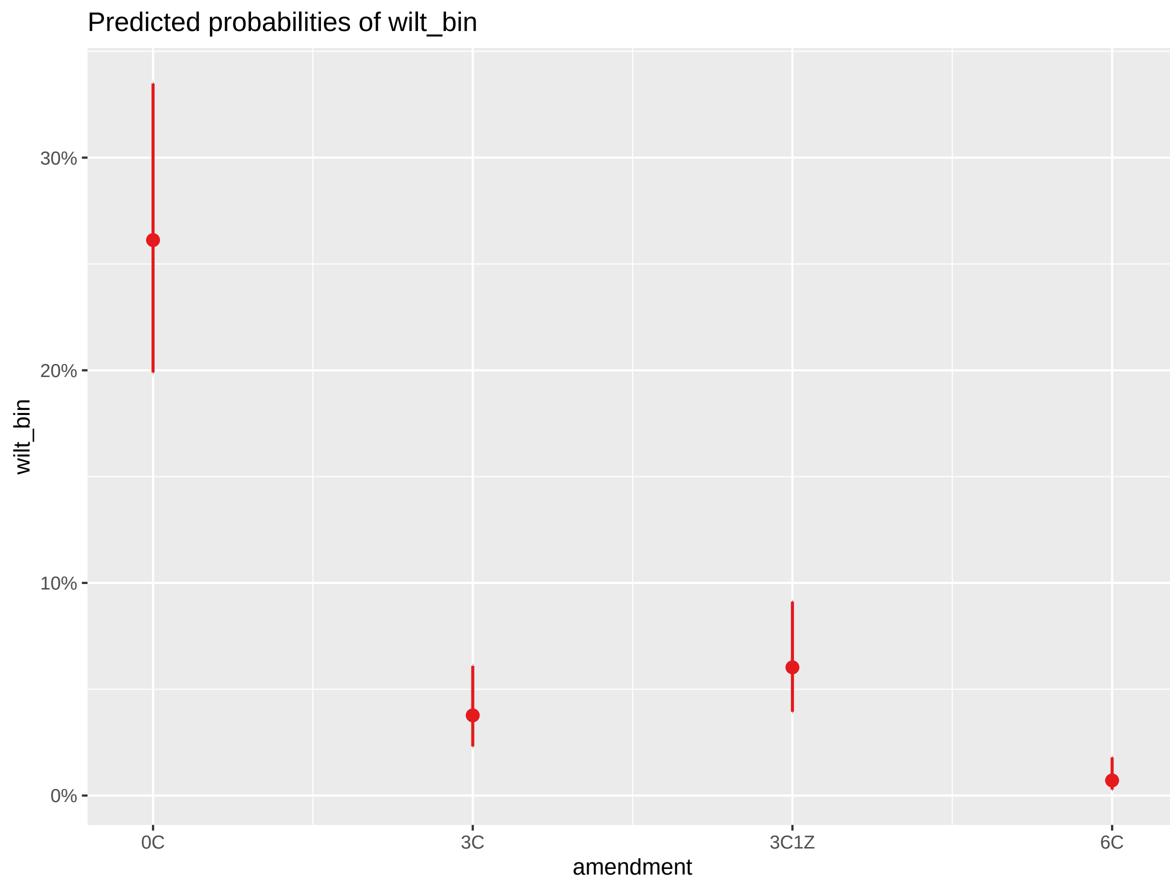
The statistical analysis consisted of a linear mixed effects model with a binary dependent variable and a blocking factor treating block as a random effect. The model was fitted using the lme4 package in R.

5 Results

The results of the analysis are presented below with the predictor names and the corresponding p-values, accompanied by a graph of the relationship between each predictor and the response variable.

```
[1] "Model Summary"

Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: wilt_bin ~ crop + amendment + (1 | block)
   Data: wilt

      AIC       BIC    logLik -2*log(L)  df.resid
   1146.5    1191.0    -565.3    1130.5      1912

Scaled residuals:
    Min      1Q  Median      3Q     Max
-0.9244 -0.3003 -0.2264 -0.0968  7.8011

Random effects:
 Groups Name        Variance Std.Dev.
 block  (Intercept) 0.003477 0.05897
Number of obs: 1920, groups:  block, 3

Fixed effects:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)    -1.0396     0.1803  -5.766 8.11e-09 ***
cropCorn        0.2867     0.2291   1.252    0.211
cropFallow      0.8484     0.2170   3.909 9.28e-05 ***
cropPea         0.1879     0.2320   0.810    0.418
amendment3C    -2.1997     0.2254  -9.761  < 2e-16 ***
amendment3C1Z  -1.7068     0.1917  -8.905  < 2e-16 ***
amendment6C    -3.8991     0.4608  -8.462  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```


Predicted probabilities of wilt_bin
50%
40%
30%
20%
wilt_bin
Cabbage
Corn
Fallow
Pea
crop



6 References

Mendiburu, Felipe de and Yaseen, Muhammad(2020). agricolae: Statistical Procedures for Agricultural Research.R package version 1.4.0, https://myaseen208.github.io/agricolae/https://cran.r-project.org/package=agricolae.

7 Appendix

Session Information

```
sessionInfo()
```

```
R version 4.5.1 (2025-06-13 ucrt)
Platform: x86_64-w64-mingw32/x64
```

```
Running under: Windows 11 x64 (build 26100)

Matrix products: default
  LAPACK version 3.12.1

locale:
[1] LC_COLLATE=English_United Kingdom.utf8
[2] LC_CTYPE=English_United Kingdom.utf8
[3] LC_MONETARY=English_United Kingdom.utf8
[4] LC_NUMERIC=C
[5] LC_TIME=English_United Kingdom.utf8

time zone: Europe/London
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
 [1] sjPlot_2.9.0    lmerTest_3.1-3  lme4_1.1-37     Matrix_1.7-3
 [5] knitr_1.50      lubridate_1.9.4 forcats_1.0.1   stringr_1.5.1
 [9] dplyr_1.1.4     purrr_1.0.4     readr_2.1.5     tidyr_1.3.1
[13] tibble_3.3.0    ggplot2_3.5.2   tidyverse_2.0.0

loaded via a namespace (and not attached):
 [1] generics_0.1.4     stringi_1.8.7      lattice_0.22-7
 [4] hms_1.1.3          digest_0.6.37      magrittr_2.0.3
 [7] evaluate_1.0.4     grid_4.5.1         timechange_0.3.0
[10] RColorBrewer_1.1-3 fastmap_1.2.0      jsonlite_2.0.0
[13] ggeffects_2.3.1    scales_1.4.0       numDeriv_2016.8-1.1
[16] reformulas_0.4.1   Rdpack_2.6.4       cli_3.6.5
[19] rlang_1.1.6        rbibutils_2.3      splines_4.5.1
[22] withr_3.0.2        yaml_2.3.10        datawizard_1.3.0
[25] tools_4.5.1        tzdb_0.5.0         nloptr_2.2.1
[28] minqa_1.2.8        sjlabelled_1.2.0   boot_1.3-31
[31] vctrs_0.6.5        R6_2.6.1           lifecycle_1.0.4
[34] MASS_7.3-65        insight_1.4.2      pkgconfig_2.0.3
[37] pillar_1.10.2      gtable_0.3.6       glue_1.8.0
[40] Rcpp_1.0.14        haven_2.5.5        xfun_0.52
[43] tidyselect_1.2.1   farver_2.1.2       htmltools_0.5.8.1
[46] nlme_3.1-168       labeling_0.4.3     rmarkdown_2.29
[49] compiler_4.5.1
```